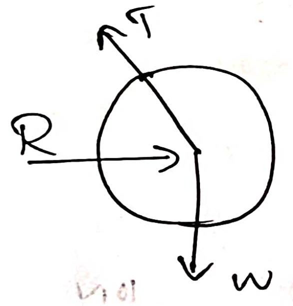
APJ Abdul Kalam Technological University
Second Semester B.Tech Degree Examination
July 2021 (2019 Scheme)

Course Code : EST 100 Course Name : Engineering Mechanics

Part A

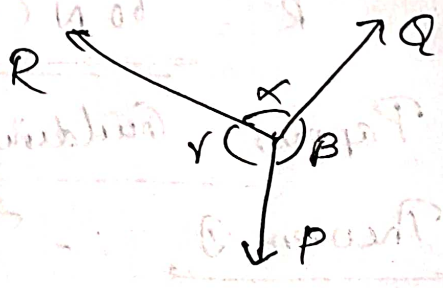
1) Free body diagram is a diagram in which a body is completely isolated from all its supports and all forces acting on the body are shown as force vectors.



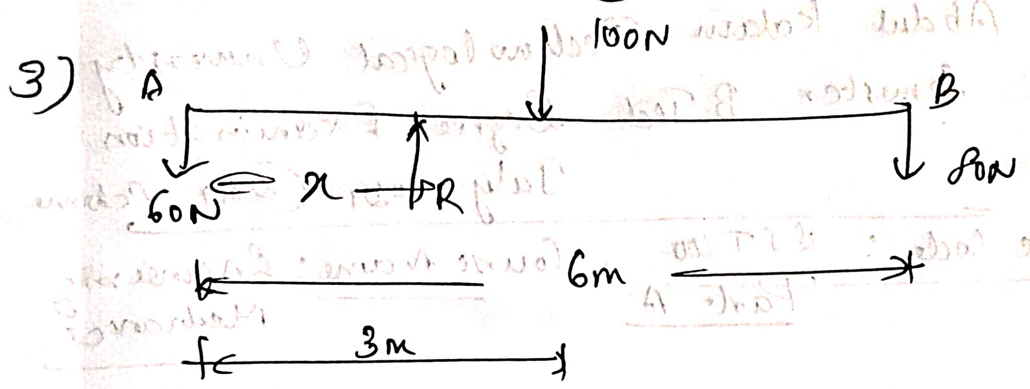
2) Lami's Theorem

If three concurrent forces are in equilibrium, then each force is directly proportional to sine of angle between ~~each~~ other two forces.

$$\frac{P}{\sin\alpha} = \frac{R}{\sin\beta} = \frac{Q}{\sin\gamma}$$



3)



$\Sigma F_y = 0$

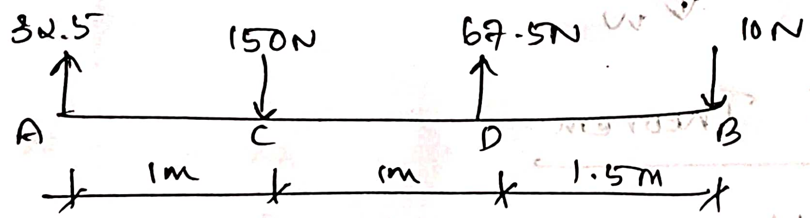
$R = 60 + 80 + 100 = 240\ N$

$\Sigma M_A = 0$

$R \cdot x = 100 \times 3 + 80 \times 6$

$x = 3.25\ m$

4)



$R = 32.5 - 150 + 67.5 - 10$

$R = -60\ N$

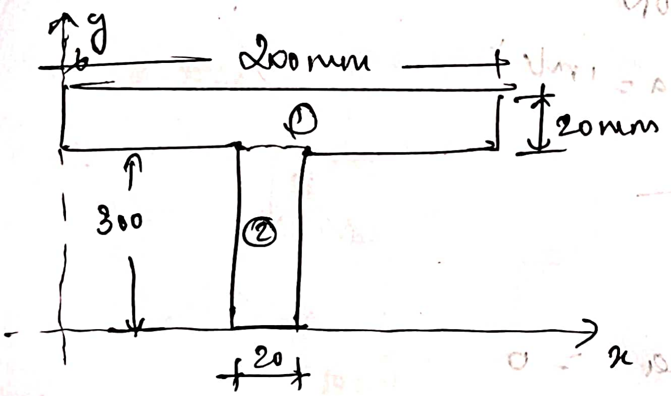
$R = 60\ N\ (\downarrow)$

5) Pappus Guldinus Theorems

Theorem ① :- The area of surface generated by revolving a plane curve about a non-intersecting axis is equal to the product of length of curve and the distance travelled by the centroid for one revolution.

Theorem ② :- The volume of solid generated by revolving a plane area about a non-intersecting axis equal to the product of area and the ~~sq~~ distance travelled by the centroid for one revolution.

6)



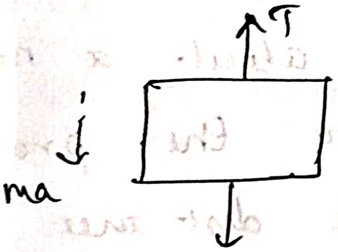
$$\bar{x} = \frac{\Sigma a_i x_i}{\Sigma a_i} \quad ; \quad \bar{y} = \frac{\Sigma a_i y_i}{\Sigma a_i}$$

$$\bar{x} = \frac{(200 \times 20) \times 100 + (20 \times 300) \times 100}{(200 \times 20) + (20 \times 300)}$$

$$\bar{x} = \underline{100 \text{ mm}}$$

$$\bar{y} = \left((200 \times 20) \times 310 + (300 \times 20) \times 150\right) / 10000$$

$$= \underline{214 \text{ mm}}$$

7) 

$a = 1 m/s^2$

$10 \times 9.81 = 98.1 N$

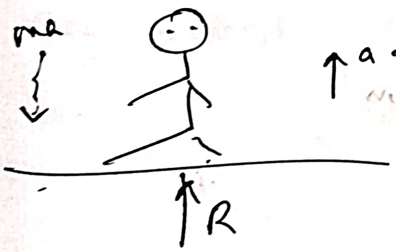
Resolving vertically,

$T - ma - 98.1 = 0$

$T - (10 \times 1) - 98.1 = 0$

$T = 108.1 N$

8) $W = 600N$



$-W + R - ma = 0$

$-600 + R - \frac{600}{(9.81)} \times 1 = 0$

$R = 661.16$

Increase in reaction $= 661.16 - 600$

$= 61.16 N$

9) Amplitude, $r = 1m$

Time Period $t_p = 2$ sec

$t_p = \frac{2\pi}{\omega} \implies \omega = \frac{2\pi}{2} = 3.14$ rad/s

$x = r \sin \omega t = 1 \times \sin\left(3.14 \times 0.4 \times \frac{180}{\pi}\right)$

$x = \underline{0.951 \text{ m}}$

$v = \omega\sqrt{r^2 - x^2} = 3.14\sqrt{1^2 - 0.951^2}$

$= 0.971 \text{ m/s}$

$a = \omega^2 \cdot x = 3.14^2 \times 0.951$

$a = \underline{9.376 \text{ m/s}^2}$

10) Instantaneous Centre ⟹ It is an imaginary point about which, the body undergoing general plane motion is assumed to be rotating for a particular instant. This point will be continuously changing. The locus of instantaneous centre is called centrode. The velocity of instantaneous centre is zero.

Velocity of any point on a rigid body performing General plane motion, is given by, $V = r.\omega$.

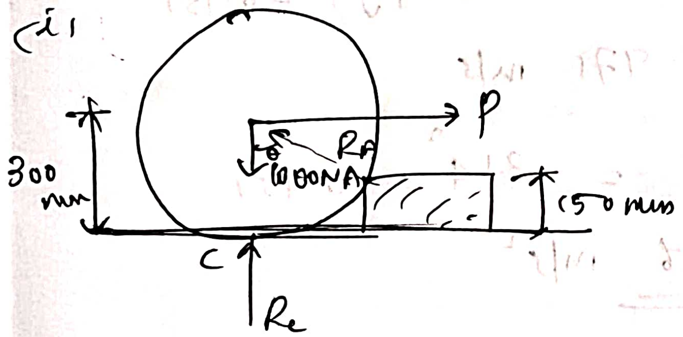
r = distance of that point from instantaneous centre

ω = angular velocity (rad/s)

Part B

(1)

(i)

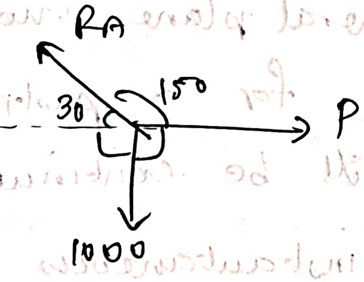


~~tan θ =~~

$\cos\theta = \frac{150}{300}$

$\theta = 60°$

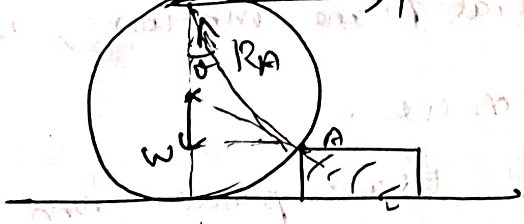
Here $R_c = 0$ as there will not be any contact as the roller moves over the block.



$$\frac{R_A}{\sin 90} = \frac{1000}{\sin 150} = \frac{P}{\sin 120}$$

$R_A = 2000\,N$, $P = 1732.05\,N$

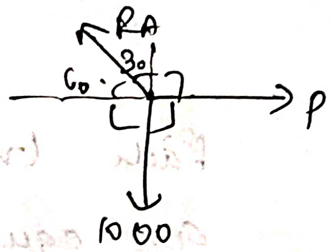
(ii)



$\tan\theta = \frac{259.8}{300+150}$

$\theta = 30°$

⑦

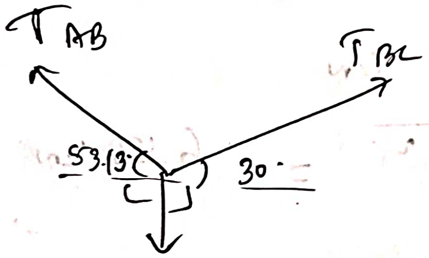


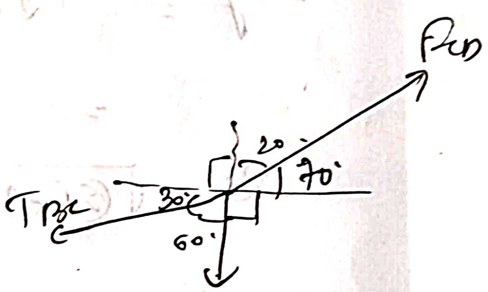
$$\frac{1000}{\sin 120} = \frac{P}{\sin 150} = \frac{R_A}{\sin 90}$$

$P = 577.35\ N$

$R_A = 1154.7\ N$

12)





$5 \times 9.81 = 49.05\ N$

$W = m \times 9.81$

$$\frac{49.05}{\sin 96.87} = \frac{T_{BC}}{\sin 143.13} = \frac{T_{AB}}{\sin 120}$$

$T_{BC} = 29.64\ N$

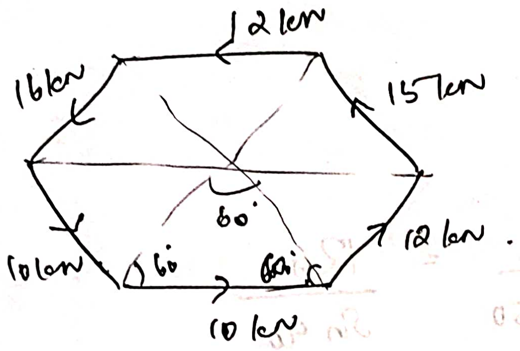
$T_{AB} = 42.785\ N$

$$\frac{F_{CD}}{\sin 60} = \frac{W}{\sin 140} = \frac{T_{BC}}{\sin 160}$$

$$F_{CD} = \frac{29.64 \times \sin 60}{\sin 160} = 75.05\ N$$

$W = 55.705\ N \implies m = 5.68\ kg$

b)



Each triangles are equilateral triangles.

$\Sigma F_x = 10 + 12\cos 60 + 10\cos 60 - 15\cos 60 - 12 - 16\cos 60 = -6.5$

$\Sigma F_y = 12\sin 60 + 15\sin 60 - 16\sin 60 - 10\sin 60$

$= 0.866$ kN

$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2} = 6.56$ kN

$\theta = 7.59°$.

Net Moment about O,

$\Sigma M_O = -(10\times 1.732) - (12\times 1.732) - (15\times 1.732) - (12\times 1.732) - (16\times 1.732) - (10\times 1.732)$

$= 129.9$ Nm

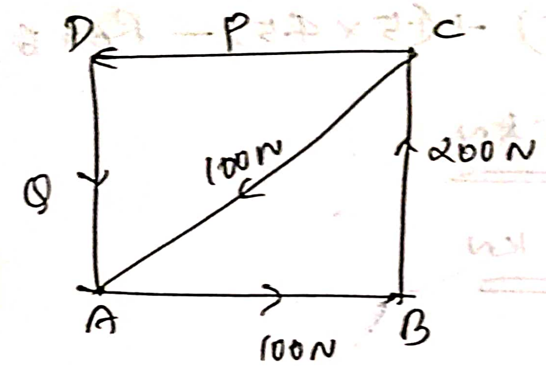
$\Sigma M = R.x$

$129.9 = 6.56 \times x$

$x = 19.8$ m (⊥r dist. from O)

Mod (1)

13) (a)



$\Sigma F_x = 0 \Rightarrow 100 - 100\cos 45 - P = 0$

$P = 29.29N$

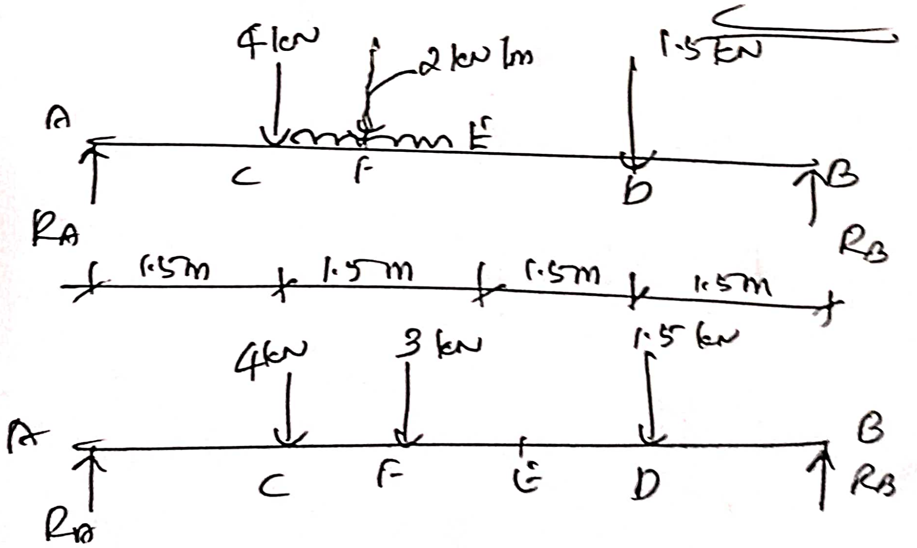
$\Sigma F_y = 0 \Rightarrow 200 - 100\sin 45 - Q = 0$

$Q = 129.29\ N$

$\Sigma M_A = -(200 \times 1) - (P \times 1)$

$= 229.29\ Nm$ (↺)

(b)



$\Sigma F_y = 0 \Rightarrow R_A + R_B = 4 + 3 + 1.5$

$R_A + R_B = 8.5\ kN$ — (1)

(10)

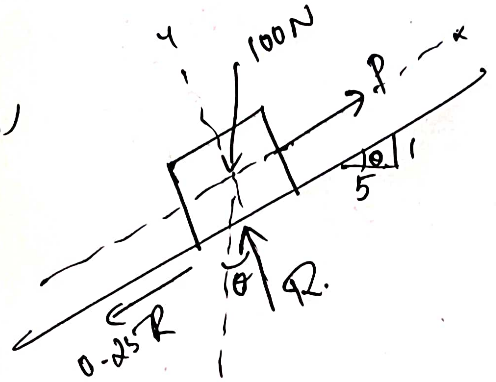
$\sum M_A = 0$

$(4 \times 1.5) + (3 \times 2.25) + (1.5 \times 4.5) - R_B \times 6 = 0$

$R_B = 3.25 \text{ kN}$

$R_A = 5.25 \text{ kN}$

14) (a)



$\tan\theta = 1/5$

$\theta = 11.53°$

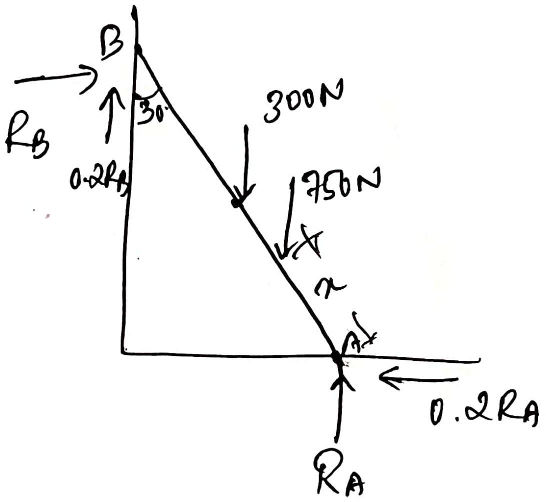
$\Sigma F_x = 0 \Rightarrow P - 100 \sin 11.53 - 0.25 R = 0$

$\Sigma F_y = 0 \Rightarrow R - 100 \cos 11.53 = 0$

$R = 97.98\ N$

$P = \underline{44.48\ N}$

(b)



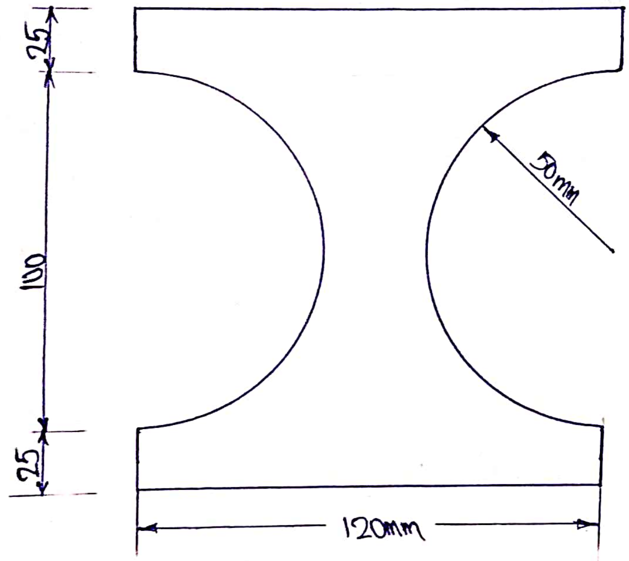
$\Sigma F_x = 0 \Rightarrow R_B - 0.2 R_A = 0$ ——①

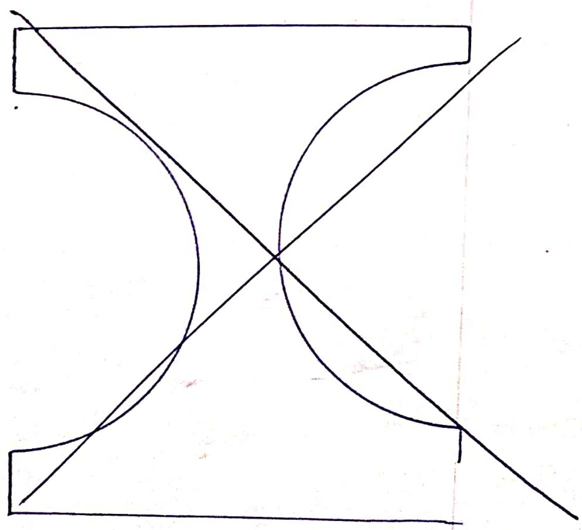
$\Sigma F_y = 0 \Rightarrow R_A + 0.2 R_B - 300 - 750 = 0$ ——②

$\Sigma M_B = 0 \Rightarrow 300 \times \frac{l}{2} \sin 30 + 750 (l - x) \sin 30 + 0.2 R_A \times l \cos 30 - R_A \times l \sin 30 = 0$

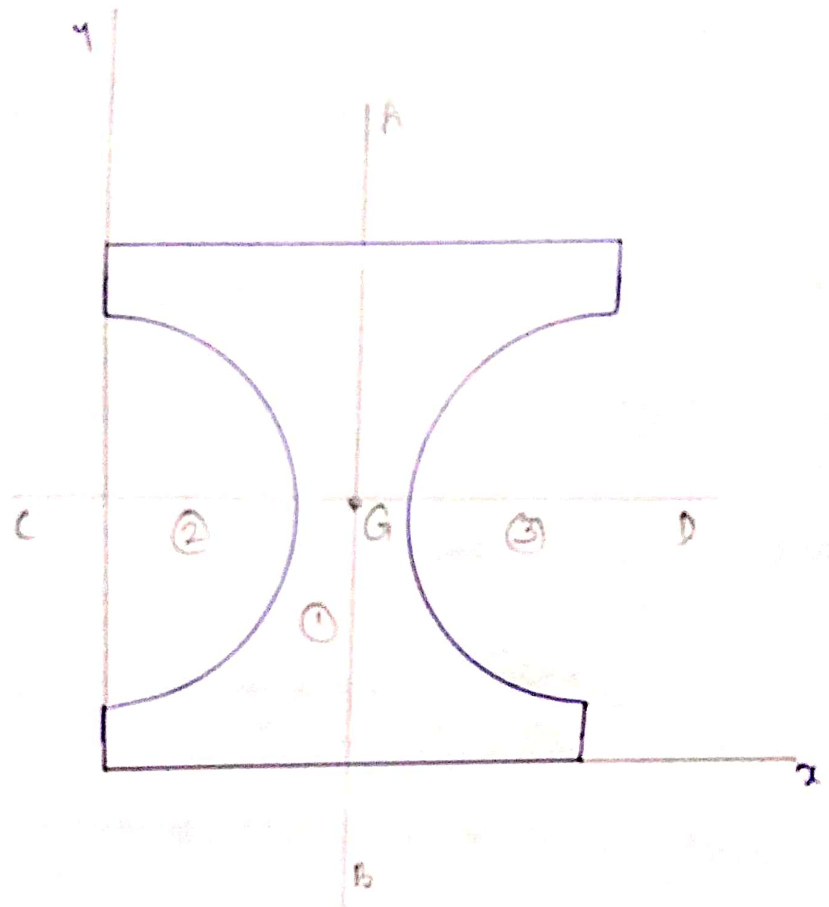
Solving, $R_A = 1009.6\ N$, $R_B = 201.9\ N$, $\underline{x = 0.321}$

15. The cross section of a cast iron beam is shown in figure. Determine the moments of inertia of the section about the horizontal and vertical axes passing through the centroid.





For finding centroid, consider the reference axes,



AB & CD axes are symmetric axes.

$\therefore \quad \bar{x} = 60mm$

$\quad \bar{y} = 75mm$

Now for finding the moment of inertia, AB and CD axes are the vertical and horizontal axes passing through centroid.

Now split the given figure into three basic figures.

= Rectangle − Semi circle − Semi circle.

i) Moment of inertia of rectangle about centroidal horizontal axis, $(I_{CD})_1$

$$(I_{CD})_1 = \frac{bd^3}{12} + 0$$

$$= \frac{120 \times 150^3}{12} = 33750000$$

ii) Moment of inertia of semicircle 2 about centroidal horizontal axis, $(I_{CD})_2$

$$(I_{CD})_2 = \frac{\pi D^4}{128} + 0$$

$$= \frac{\pi \times 100^4}{128} = 2454369.26$$

iii) Moment of inertia of semicircle 3 about centroidal horizontal axis, $(I_{CD})_3$

$$(I_{CD})_3 = \frac{\pi D^4}{128} + 0$$

$$= \frac{\pi \times 100^4}{128} = 2454369.26$$

Moment of inertia of given cross section about centroidal horizontal axis,

$$I_{CD} = (I_{CD})_1 - (I_{CD})_2 - (I_{CD})_3$$

$$= 28841261.5 = \underline{28.84 \times 10^6 \, mm^4}$$

Moment of inertia of cross section about centroidal vertical axis,

i) Moment of inertia of rectangle about centroidal vertical axis,

$$(I_{AB})_1 = \frac{db^3}{12} + 0$$

$$= \frac{120^3 \times 150}{12} = 21600000$$

ii) Moment of inertia of semicircle 2 about centroidal vertical axis,

$$(I_{AB})_2 = 0.11R^4 + A_2 \times (d_x)^2$$

$$= \left[0.11 \times 50^4\right] + \left[\frac{\pi \times 50^2}{2} \times \left(60 - \frac{4 \times 50}{3\pi}\right)^2\right]$$

$$= \left[0.11 \times 50^4\right] + \left[\frac{\pi \times 50^2}{2} \times 38.78^2\right]$$

$$= 687500 + \cancel{3252680.94}\ 5905755.95$$

$$= \cancel{3940180.94}\ 6593255.95$$

iii) Moment of inertia of semicircle 3 about centroidal vertical axis,

$$(I_{AB})_3 = 0.11R^4 + A_3 (d_x)^2$$

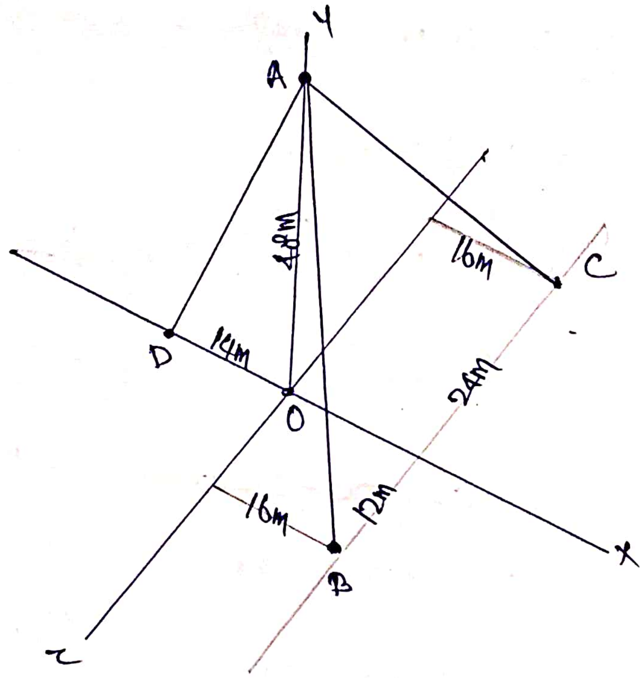
$$= \left[0.11 \times 50^4\right] + \left[\frac{\pi \times 50^2}{2} \times \left(60 - \frac{4 \times 50}{3\pi}\right)^2\right]$$

$$= \cancel{3940180.94}\ 6593255.95$$

$$I_{AB} = (I_{AB})_1 - (I_{AB})_2 - (I_{AB})_3$$

$$= 8413488.1 = \underline{\underline{8.41 \times 10^6\ mm^4}}$$

16. A post is held in vertical position by three cables AB, AC and AD as shown in figure. If the tension in cable AB is 40N, calculate the tension in AC and AD so that the resultant of three forces at A is vertical.



Point A $(0, 48, 0)$

B $(16, 0, 12)$

C $(16, 0, -24)$

D $(-14, 0, 0)$

We know forces are acting along AB, AC and AD direction.

Force along AB is $\vec{F}_{AB} = |\vec{F}_{AB}| \lambda_{AB}$

$$\lambda_{AB} = \frac{\vec{AB}}{|\vec{AB}|} = \frac{(16-0)\hat{i}\ (0-48)\hat{j}\ (12-0)k}{\sqrt{16^2 + -48^2 + 12^2}}$$

$$= \frac{16\hat{i} - 48\hat{j} + 12k}{\sqrt{2704}} = \frac{16\hat{i} - 48\hat{j} + 12k}{52}$$

$$\vec{F_{AB}} = |F_{AB}| \cdot \frac{16\hat{i} - 48\hat{j} + 12k}{52} = \frac{40}{52}(16\hat{i} - 48\hat{j} + 12k)$$

Similarly

$$\vec{F_{AC}} = |\vec{F_{AC}}| \cdot \frac{16\hat{i} - 48\hat{j} - 24k}{56}$$

$$F_{AD} = |\vec{F_{AD}}| \cdot \frac{-14\hat{i} - 48\hat{j} + 0k}{50}$$

Now we know resultant of these three forces at A is vertical. That mean it will have only y component.

$$\vec{R} = \vec{F_{AB}} + \vec{F_{AC}} + \vec{F_{AD}}$$

$$= \frac{40}{52}(16\hat{i} - 48\hat{j} + 12k) + |\vec{F_{AC}}|\left(\frac{16\hat{i} - 48\hat{j} - 24k}{56}\right) + |\vec{F_{AD}}|\left(\frac{-14\hat{i} - 48\hat{j}}{50}\right)$$

The y component of this resultant vector is not equal to zero. but other two components are zero. i.e. $R_x = R_z = 0$

$$R_x = \frac{40}{52} \times 16\hat{i} + |\vec{F_{AC}}| \cdot \frac{16}{56} + |\vec{F_{AD}}| \cdot \frac{-14}{50} = 0$$

$$= 12.30 + 0.285|\vec{F_{AC}}| - 0.28|\vec{F_{AD}}| = 0 \quad \text{——— (1)}$$

$$R_x = \frac{40}{52} \times 12 + |\vec{F_{AC}}| \cdot \frac{-24}{56} + 0 = 0$$

$$= 9.23 - 0.428\,|\vec{F_{AC}}| = 0 \quad \text{——— (2)}$$

From eq. (2)

$$\Rightarrow 0.428\,|\vec{F_{AC}}| = 9.23$$

$$|\vec{F_{AC}}| = \underline{\underline{21.56\,N}}$$

Substitute F_{AC} in eq. (1)

$$\Rightarrow 12.30 + 0.285 \times 21.56 - 0.28\,|\vec{F_{AD}}| = 0$$

$$= 0.28\,|\vec{F_{AD}}| = 18.44$$

$$|\vec{F_{AD}}| = \frac{18.44}{0.28} = \underline{\underline{65.87\,N}}$$

Force along AC = $\underline{\underline{21.56N}}$

AD = $\underline{\underline{65.87\,N}}$

17.a.

In a motion of a projectile, in what proportion will the maximum range be increased if the initial velocity is increased by 10%.

Range, $R = \frac{u^2 \sin 2\theta}{g}$

$R_{max} = \frac{u^2}{g}$

If initial velocity increased by 10%.

$u = 1.1u$

Then $(R_{max})' = \frac{(1.1u)^2}{g} = \frac{1.21u^2}{g} = 1.21 R_{max}$

That indicates, maximum range will increase by 21%.

b. A train weighing 1700kN without the engine starts to move with constant acceleration along a straight line track and in the first 60 seconds acquires a velocity of 54km/hr. Determine the tension in the coupling between the train & the engine if the total resistance to motion due to friction and air resistance is constant and equal to 0.005 times the weight of the train.

Given:-

$W = 1700kN = 1700 \times 10^3 N$

$m = \frac{1700 \times 10^3}{9.81} kg$

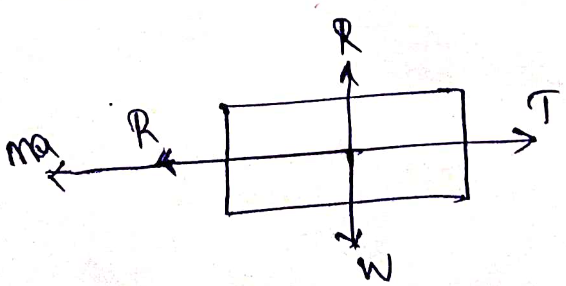
$t = 60 \text{ sec}$

$V = 54 km/hr = 54 \times \frac{5}{18} = 15 m/s$

$V = u + at$

$15 = 0 + a \times 60 \Rightarrow a = \cancel{60}\ 15/60 = \underline{0.25 m/s^2}$

Now consider FBD



W → Weight
R → Normal reaction
T → Tension in coupling
R → Total resistance
ma → Inertia force

$\Sigma F_x = 0$

$T = R + ma$

$= 0.005 \times 1700 \times 10^3 + \frac{1700 \times 10^3}{9.81} \times 0.25$

$= 8500 + 43323.13$

$= \underline{\underline{51823.13N}}$

18. a. A ball of mass 'm' is dropped from rest from the top of a tower of height H. Write the equations of kinematics for the motion of ball under free fall at any instant 't' of motion.

$u = 0$

$v = u + at \Rightarrow v = gt$

$S = ut + \frac{1}{2}at^2 \Rightarrow H = \frac{1}{2}gt^2$

$v^2 = u^2 + 2as \Rightarrow v^2 = 2gH$

18. b. Three spherical balls of mass 2kg, 6kg and 12kg are moving in the same direction with velocities 12m/s, 4m/s and 2m/s respectively. If the ball of mass 2kg impinges with the ball of mass 6kg, which in turn impinges with the ball of mass 12kg, prove that the balls of masses 2kg and 6kg will be brought to rest by the impacts. Assume to be perfectly elastic.

OUT OF SYLLABUS.

19. An inextensible rope passing over a smooth pulley has two blocks of mass 20kg and 30kg attached to its two ends. The mass of the pulley is 10kg and radius of gyration 0.3m. Determine the tension on the rope and acceleration of the masses.

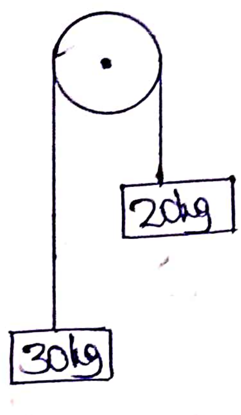
<u>Given:-</u>

$m_1 = 20kg$

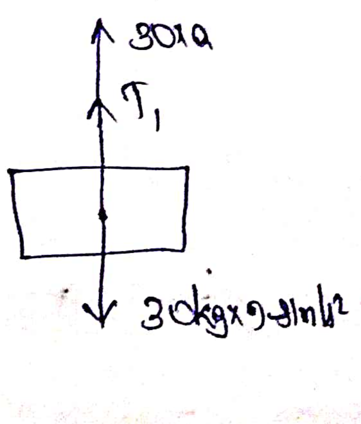
$m_2 = 30kg$

$m_p = 10kg$

$k = 0.3m$

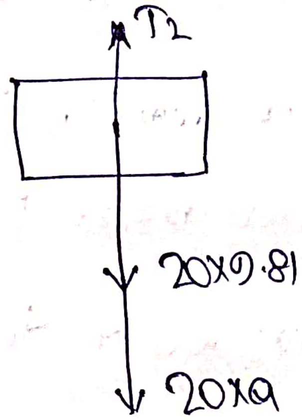


Consider 30kg block



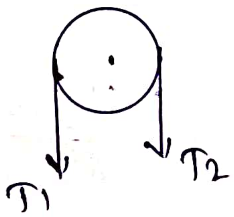
$T_1 = 30\times9.81 - 30a$ —— (1)

Consider the 20kg block



$$T_2 = 20\times9.81 + 20a \quad \text{---(2)}$$

Consider the pulley



$$I\alpha = T_1.r - T_2.r$$

$$I = mk^2 = 10\times0.3^2$$

$$= \cancel{3\,kgm^2} = 0.9\,kgm^2 \qquad \Rightarrow I = \frac{mr^2}{2} \Rightarrow r = 0.42m$$

$$\alpha = ar$$

Now, $I\alpha = T_1.r - T_2.r$

$$0.9\,\cancel{3\times\frac{a}{r}} = [30\times9.81 - 30a] - [20\times9.81 + 20a]\,ar$$

$$0.9a\,\cancel{3a} = 30\times9.81 - 20\times9.81 - 30a - 20a$$

$$0.9 \cdot \frac{a}{r} = [30\times 9.81 - 30a]r - [20\times 9.81 + 20a]\times r$$

$$0.9a = [30\times 9.81 - 30a - 20\times 9.81 - 20a]\times r^2$$

$$0.9a = (98.1 - 50a)\times 0.424^2$$

$$0.9a = 17.63 - 8.98a$$

$$9.88a = 17.63$$

$$a = \underline{\underline{1.78\ m/s^2}}$$

$$T_1 = 30\times 9.81 - 30\times 1.78$$

$$= \underline{\underline{240.9N}}$$

$$T_2 = 20\times 9.81 + 20\times 1.78$$

$$= \underline{\underline{231.8N}}$$

20. In a particular SHM performed by a particle of mass m, the amplitude is 1.57m and time period of oscillation is 5s.

i) Calculate velocity and acceleration of particle at ~~0.5m~~ 0.53m away from centre

ii) Determine magnitude and location of maximum velocity and maximum acceleration of particle.

iii) Also determine the time required by the particle to pass two points 1.35m away and 0.53m away from the central point of oscillation. Both the points lie on the same side of central point.

Given:-

$r = 1.57m$

$T = 5 sec$

i) ~~$x = 1.07m$~~

$x = 0.53m$

$$V = \omega\sqrt{r^2 - x^2}$$

$$a = -\omega^2 . x$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{5} \text{ rad/s}$$

$$V = \frac{2\pi}{5}\sqrt{1.57^2 - 0.53^2} = \underline{\underline{1.85 \text{ m/s}}}$$

$$a = -\omega^2 x = -\left(\frac{2\pi}{5}\right)^2 \times 0.53 = \underline{\underline{0.83 \text{ m/s}^2}}$$

ii) Maximum velocity

Maximum velocity will be at mean position

$$V_{max} = \omega r$$

$$= \frac{2\pi}{5} \times 1.57$$

$$= \underline{\underline{1.97 \text{ m/s}}}$$

Maximum acceleration

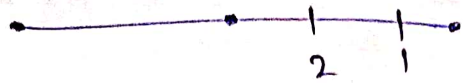
Acceleration will be maximum at extreme position

$$a_{max} = -\omega^2 r$$

$$= \frac{4\pi^2}{25} \times 1.57$$

$$= \underline{\underline{2.47 \text{ m/s}^2}}$$

iii)



Consider the point 1, ie 1.35m away from center

$\Rightarrow x_1 = 1.35m$

Similarly $x_2 = 0.53m$

We know, $x = r \cos \omega t$

$$x_1 = 1.57 \cos(\omega t_1)$$

$$1.35 = 1.57 \cos\left(\frac{2\pi}{5} \times t_1\right)$$

$$0.859 = \cos\left(\frac{2\pi}{5} \times t_1\right)$$

$$\frac{2\pi}{5} \times t_1 = 30.79^\circ$$

$$\frac{2\pi}{5} \times \frac{180}{\pi} \times t_1 = 30.79^\circ$$

$$t_1 = \underline{0.427 \text{ sec}}$$

Similarly, $x_2 = 1.57 \cos(\omega t_2)$

$$0.53 = 1.57 \cos\left(\frac{2\pi}{5} \times t_2\right)$$

$$0.33 = \cos\left(\frac{2\pi}{5} \times t_2\right)$$

$$\frac{2\pi}{5} t_2 = 70.27^\circ$$

$$\Rightarrow \frac{2\pi}{5} \times \frac{180}{\pi} \times t_2 = 70.27^\circ$$

$$t_2 = \underline{0.975 \text{ sec}}$$

Time required by the particle to pass between two points $= t_2 - t_1$

$$= 0.975 - 0.427$$

$$= \underline{0.548 \text{ s}}$$